What ReqShift does, one stage at a time

The six stages a run passes through, and what each one is for.

The whole idea in one line. A language model cannot tell you when it is wrong, and a solver cannot read prose. So the model reads and proposes, and everything after that exists to check it. Only one of the six stages guesses anything.

1 Reading the requirements

read

The text is pulled out of the PDF and the things PDFs break are repaired: page headers that repeat on every page, words split across a line end, sentences wrapped over several lines. This matters because every finding later has to point back at a real sentence in your document, and a sentence broken in half cannot be quoted or checked.

2 Checking the document is in scope

done

One cheap question before any money is spent: *does this document say how many of each thing there can be?* That takes about ten seconds, against twenty minutes to discover the same thing by extracting first. A document that never states a quantity has nothing for the analysis to combine, and is refused with the reason given.

3 AI drafts a goal model

model drafted

The only stage where anything is guessed. The model reads the whole document and proposes the participants, what each one has to achieve, how those goals break into sub-goals, who depends on whom, and how many of each there can be. Everything after this exists to check this step. It is a draft, and it is labelled as one wherever it appears.

4 Checking the model is well formed

well formed

No AI here, only structure. Does every goal have an owner, does every participant have an overall goal, does anything point at something that is not there. A malformed draft is caught now rather than after nine audits have been paid for.

5 Auditing what each number counts

the long one

Nine separate passes, each asking one narrow question, each a fresh call with no memory of the extraction so it is not anchored to the first answer. This is where most of the time and most of the cost goes.

- Does that sentence really state this maximum, or does the number count something else? “The equipment shall have six output ports” bounds ports, not clocks.
- Can there be **zero** of this, or did the analysis assume so? That one field decides whether there is a finding at all.
- Are the capacity, demand and timing numbers real?
- Does each dependency actually exist in the text, or is it merely plausible?
- Are the AND/OR breakdowns and the overall goals supported by a sentence?
- What did the extraction **miss**? Every other pass is blind to an omission, and an omission turns a real defect into a clean bill of health.

An audit is an improvement to the model, never a gate on it. One that cannot finish costs its own contribution and nothing else, and the model is saved after each pass so no failure can waste the work already done.

6 The solver proves what follows

z3, or GoalShift

The audited model goes to the solver, which works out every build the bounds allow, checks each one, and groups them into distinct behaviours. Nothing is sampled and nothing is summarised.

If the banner says **z3** rather than GoalShift, the model stated durations or capacities, which the SAT encoding would silently ignore rather than use. The engine follows the model rather than a dropdown, and the result then carries the caveat that it is not proved by the encoding the thesis describes. That is honesty, not an error.

What comes back

First the gap report: the quantities the document never states, ranked by whether their absence can actually break something, each with the sentence it was read from. Every entry reports an absence, which you can check by searching your own document.

Then the findings: which builds cannot work, and which participant is present and provably unable to do its job.

> **Read the marks.** A finding that rests on something the document does not state is labelled **unconfirmed** and carries the sentence it depends on. The solver result is correct for the model as drawn. Whether the model is right about your specification is the part still open, and the report says which is which rather than counting them together.